

Topological Corroboration via Simplicial Closures for Heterophilic Multimodal Graph Clustering

Anonymous submission

Abstract

Recent advances in multimodal graph clustering disentangle homophilic and heterophilic signals into low and high frequency channels. Yet the topological context of heterophilic evidence remains underexplored. Low frequency representations preserve cluster interiors, whereas pairwise high frequency variation is structurally underdetermined: discrepancy alone cannot distinguish isolated noise from a boundary supported by local organization. Independent modality encoding further overlooks closures jointly supported by edges across modalities, while relation fusion that ignores frequency channels overlooks relation utility that depends on frequency. We introduce Topological Corroboration via Simplicial Closures (TopoCor), which advances heterophily modeling by contextualizing pairwise high frequency discrepancies through simplicial closure structure. TopoCor preserves low frequency context as a stable cluster anchor and lifts the high frequency support in each modality into a clique complex. Filters for each interaction order distinguish pairwise edges from triangular closures, while an orthogonal residual update injects closure evidence complementary to the pairwise anchor. Relation-Frequency Decoupled Calibration (RFDC) aligns relation contributions with their respective roles in the low and high frequency channels. TopoCor further constructs closures jointly supported across modalities and routes their evidence according to edge provenance, making topology across modalities explicit before representation fusion. Across six multimodal benchmarks with multiple relations, TopoCor ranks first in all 18 dataset-metric comparisons across ACC, NMI, and ARI.

1 Introduction

Multimodal graphs couple heterogeneous attributes with relational structure, and clustering seeks latent organization shared across these sources (Ektefaie et al. 2023; Zhu et al. 2025; Wang et al. 2025a; Guo et al. 2025; Zheng et al. 2026; Xu et al. 2022; Zhang et al. 2019). Existing methods integrate graph topology with multimodal attributes (Wang et al. 2025a; Jia et al. 2023; He et al. 2025), model modality correlation and feature noise (Zheng et al. 2026), or separate homophilic and heterophilic signals (Guo et al. 2025; Zhu et al. 2020; Luan et al. 2022; Li et al. 2022). Yet models that disentangle frequency channels still encode candidate heterophily mainly through pairwise high frequency variation. Such variation identifies discrepancy without establishing

structural support.

This evidence is therefore *structurally underdetermined*: endpoint discrepancy alone cannot tell isolated noise from a boundary supported by local organization (Wang et al. 2024; Zheng, Luan, and Chen 2024; Zhu et al. 2020; Lim et al. 2021). Higher order topology supplies the missing context. Simplicial closure records how lower order faces jointly support a higher order simplex (Benson, Gleich, and Leskovec 2016; Benson et al. 2018; Bodnar et al. 2021; Feng et al. 2019), and this incidence is not recovered by deepening pairwise propagation: multi-hop aggregation summarizes walk connectivity (Gasteiger, Bojchevski, and Günnemann 2019; Abu-El-Haija et al. 2019; Chien et al. 2021), whereas triangle co-membership encodes shared closure participation. We define *topological corroboration* as using such closure structure to contextualize pairwise discrepancy rather than to replace it. Figure 1(a,b) shows both the structural distinction and its effect on high frequency representations.

This gap becomes more consequential across modalities. Existing graph models encode or aggregate evidence from individual modalities without constructing closures jointly supported by their edges (Jia et al. 2023; Wang et al. 2025a; Guo et al. 2025; Tao et al. 2020; He et al. 2025). Concatenation and attention at the representation level can mix or reweight unimodal evidence (Gong, Nie, and Xu 2022; Yan et al. 2023; Zheng et al. 2026), but they do not explicitly construct closures absent from any single modality or retain the edge provenance needed to attribute them. This motivates constructing closure structure across modalities before representation fusion. Figure 1(c) illustrates this gap.

Multiple relations introduce a separate assembly problem. Relation views differ in semantics, redundancy, and density (Mo et al. 2023; Shen, Wang, and Kang 2024; Shen, He, and Kang 2024; Jing, Park, and Tong 2021), yet conventional fusion assigns each relation a single global importance. This assumption is restrictive because the same relation may support smooth cluster interiors but contribute little to non-smooth boundaries, or vice versa. Averaging erases this polarity, while a shared relation gate cannot express it. Relation utility should therefore be calibrated separately across frequency channels before corroboration across modalities.

TopoCor addresses these gaps by moving heterophily modeling from pairwise discrepancy detection to topological corroboration. Contextualized Frequency Decomposition first

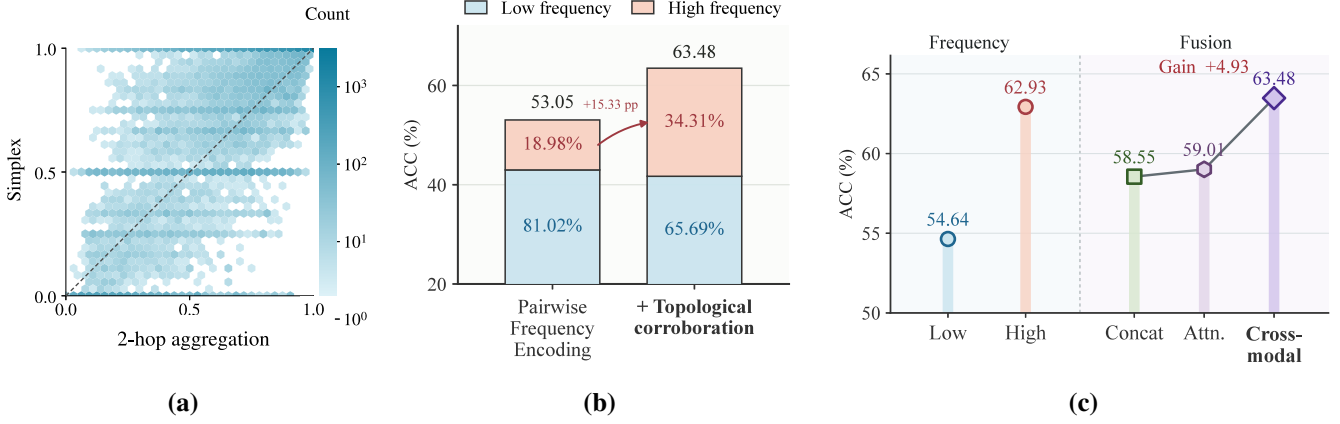


Figure 1: Empirical motivation on Ele-Fashion. (a) Aggregation over multiple hops differs from simplex closure responses. (b) Encoding with closure structure improves high frequency utilization. (c) Encoding by modality followed by representation fusion omits explicit closure structure across modalities.

separates high frequency anchors specific to each modality from shared low frequency context. Within-modal Simplicial Corroboration (WMSC) then models edge and triangle organization with responses specific to each order and injects closure evidence complementary to each pairwise anchor (He et al. 2021; Bodnar et al. 2021). Relation-Frequency Decoupled Calibration (RFDC) subsequently aggregates representations specific to each relation using weights that depend on frequency. Cross-modal Closure Corroboration (CMCC) finally constructs closures from high frequency edge sets in each modality and routes their evidence to the resulting anchors at the modality level according to edge provenance. Together, these operations preserve low frequency cluster geometry while corroborating candidate heterophilic evidence within and across modalities.

Our contributions are threefold:

- We formulate *topological corroboration* for heterophilic multimodal graph clustering, recasting pairwise discrepancy as evidence to be contextualized by simplicial closures rather than treated as a pairwise detection signal alone.
- We develop a corroboration framework at two levels in which WMSC encodes closure structure specific to each interaction order within modalities, while CMCC constructs closures across modalities and routes their evidence according to edge provenance. A shared residual update preserves pairwise anchors and injects complementary closure evidence.
- We introduce Relation-Frequency Decoupled Calibration (RFDC) to preserve relation utility specific to each frequency channel before corroboration across modalities. Across six benchmarks, TopoCor ranks first in all 18 dataset–metric comparisons, while targeted ablations isolate the roles of its core mechanisms.

2 Methodology

Overview. TopoCor contextualizes pairwise heterophily through *topological corroboration* (Figure 2). Contextu-

alized Frequency Decomposition builds shared low frequency interiors and modality-specific high frequency anchors; WMSC corroborates each anchor with order-specific closure evidence; RFDC calibrates relation views separately in each frequency channel; and CMCC routes cross-modal closure evidence by edge provenance before training.

2.1 Problem Formulation

We study unsupervised clustering on an attributed multi-graph

$$\mathcal{G} = (\mathcal{V}, \{\mathbf{X}^{(m)}\}_{m=1}^M, \{\mathbf{A}^{(r)}\}_{r=1}^R), \quad (1)$$

where $\mathcal{V} = \{v_i\}_{i=1}^N$, $\mathbf{X}^{(m)} \in \mathbb{R}^{N \times d_m}$ are modality attributes, and $\mathbf{A}^{(r)} \in \{0, 1\}^{N \times N}$ are relation adjacencies; $M=1$ and $R=1$ recover multi-relational and multimodal graphs. Given $\mathcal{G}$ and cluster number K , we learn $\mathbf{Z} \in \mathbb{R}^{N \times d}$ from which K -means recovers a partition of $\mathcal{V}$.

2.2 Contextualized Frequency Decomposition

Starting from the raw modality attributes $\mathbf{X}^{(m)}$, we first contextualize each modality on every relation while retaining a restart to the original features. Let $\mathbf{D}^{(r)}$ be the degree matrix of $\mathbf{A}^{(r)}$ and write $\hat{\mathbf{A}}^{(r)} = (\mathbf{D}^{(r)})^{-\frac{1}{2}} \mathbf{A}^{(r)} (\mathbf{D}^{(r)})^{-\frac{1}{2}}$ for its symmetrically normalized adjacency. Personalized PageRank with restart probability α_{ppr} produces relation-conditioned features

$$\begin{aligned} \mathbf{X}_0^{(m,r)} &= \mathbf{X}^{(m)}, \\ \mathbf{X}_{t+1}^{(m,r)} &= (1 - \alpha_{\text{ppr}}) \hat{\mathbf{A}}^{(r)} \mathbf{X}_t^{(m,r)} + \alpha_{\text{ppr}} \mathbf{X}^{(m)}. \end{aligned} \quad (2)$$

After T steps we set $\mathbf{X}^{(m,r)} = \mathbf{X}_T^{(m,r)}$. A view-specific linear map $\mathbf{W}^{(m,r)} \in \mathbb{R}^{d_m \times d}$ then aligns widths,

$$\mathbf{H}^{(m,r)} = \mathbf{X}^{(m,r)} \mathbf{W}^{(m,r)} \in \mathbb{R}^{N \times d}, \quad (3)$$

so every subsequent low-/high-frequency computation consumes the same projected features that already carry both modality content and relation context.

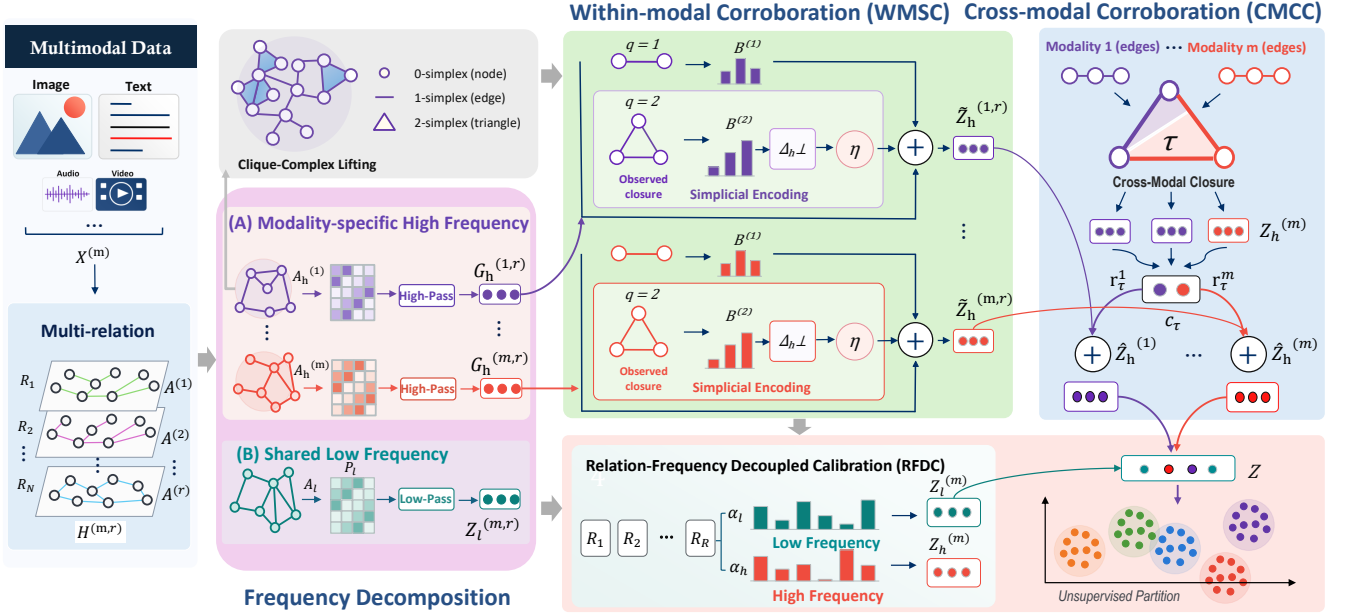


Figure 2: Overview of TopoCor. Frequency decomposition produces structural anchors, WMSC corroborates each high frequency anchor within its modality, RFDC aggregates representations specific to each relation with weights that depend on frequency, and CMCC routes closure evidence across modalities before hierarchical fusion.

We next construct graph operators that separate smoothing from variation. Let $\mathbf{Y}^{(m)}$ be the row-normalized concatenation of $\{\mathbf{H}^{(m,r)}\}_{r=1}^R$ and define the similarity $\mathbf{S}^{(m)} = \text{RowMinMax}(\mathbf{Y}^{(m)} \mathbf{Y}^{(m)\top})$, evaluated blockwise so that only the top- k entries of each row are retained. A consensus top- k_ℓ sparsification of $\{\mathbf{S}^{(m)}\}_{m=1}^M$ produces the shared homophilic support $\mathbf{A}_\ell$; for each modality, top- k_h selection on $\mathbf{I} - \mathbf{S}^{(m)}$ produces the heterophilic support $\mathbf{A}_h^{(m)}$, with edge set $\mathcal{E}_h^{(m)}$. The corresponding operators are

$$\begin{aligned} \mathbf{P}_\ell &= \mathbf{D}_\ell^{-\frac{1}{2}} \mathbf{A}_\ell \mathbf{D}_\ell^{-\frac{1}{2}}, \\ \mathbf{P}_h^{(m)} &= \mathbf{I} - (\mathbf{D}_h^{(m)})^{-\frac{1}{2}} \mathbf{A}_h^{(m)} (\mathbf{D}_h^{(m)})^{-\frac{1}{2}}, \end{aligned} \quad (4)$$

where $\mathbf{D}_\ell$ and $\mathbf{D}_h^{(m)}$ are the corresponding degree matrices. A shared encoder then returns the low-frequency interior and the pairwise high-frequency anchor

$$\begin{aligned} \mathbf{Z}_\ell^{(m,r)} &= \text{Enc}(\mathbf{H}^{(m,r)}, \mathbf{P}_\ell), \\ \mathbf{G}_h^{(m,r)} &= \text{Enc}(\mathbf{H}^{(m,r)}, \mathbf{P}_h^{(m)}). \end{aligned} \quad (5)$$

Here $\mathbf{Z}_\ell^{(m,r)}$ preserves smooth cluster geometry, while $\mathbf{G}_h^{(m,r)}$ retains endpoint variation. The latter remains undetermined until local closures establish whether that variation participates in a coherent group.

2.3 Within-modal Simplicial Corroboration (WMSC)

Simplicial lifting. Pairwise discrepancy on $\mathbf{A}_h^{(m)}$ does not reveal whether an edge participates in a shared local configuration. We therefore lift the high-frequency support to its

clique complex $\mathcal{K}_h^{(m)} = (\mathcal{V}, \mathcal{E}_h^{(m)}, \mathcal{T}_h^{(m)})$, where $\mathcal{E}_h^{(m)}$ and $\mathcal{T}_h^{(m)}$ contain, respectively, the edges of $\mathbf{A}_h^{(m)}$ and all triangles closed in that support; no edge or simplex is synthesized. Let $\mathbf{B}_q^{(m)} \in \{0, 1\}^{N \times |\mathcal{S}_q^{(m)}|}$ be the unsigned node-to- q -simplex incidence matrix for $q \in \{1, 2\}$, with $\mathcal{S}_1^{(m)} = \mathcal{E}_h^{(m)}$ and $\mathcal{S}_2^{(m)} = \mathcal{T}_h^{(m)}$. The normalized co-membership operator

$$\mathbf{P}_{\text{RW}}^{(m,q)} = \frac{1}{q+1} (\mathbf{D}_q^{(m)})^{-\frac{1}{2}} \mathbf{B}_q^{(m)} \mathbf{B}_q^{(m)\top} (\mathbf{D}_q^{(m)})^{-\frac{1}{2}} \quad (6)$$

aggregates features over simplices of order q , where $\mathbf{D}_q^{(m)}$ stores node-simplex degrees. Order $q = 1$ encodes edge incidence; $q = 2$ requires a closed triangle.

This motivates separating edge and triangle channels rather than deepening pairwise propagation. For a simple undirected graph with adjacency $\mathbf{A}$ and unsigned triangle incidence $\mathbf{B}_2$, write $\mathbf{C}_2 = \mathbf{B}_2 \mathbf{B}_2^\top$.

Proposition 1 (Closure walk separation) *The triangle co-membership operator satisfies*

$$\mathbf{C}_2 = \mathbf{A} \odot \mathbf{A}^2 + \frac{1}{2} \text{Diag}(\text{diag}(\mathbf{A}^3)). \quad (7)$$

Consequently, no scalar polynomial p satisfies $\mathbf{C}_2 = p(\mathbf{A})$ for every simple undirected graph.

Thus closure incidence is distinct from walk reachability (Figure 1(a)); details appear in the supplementary material.

Order-specific responses and residual corroboration. We therefore learn an independent Bernstein response of

degree J on each order (He et al. 2021):

$$\mathcal{H}_q(\mathbf{H}) = \sum_{p=0}^J c_{q,p} \binom{J}{p} \mathbf{P}^p (\mathbf{I} - \mathbf{P})^{J-p} \mathbf{H}, \quad \mathbf{P} = \mathbf{P}_{\text{RW}}^{(m,q)}, \quad (8)$$

with monotone cumulative softplus coefficients $c_{q,p}$. Applying $\mathcal{H}_q$ to the projected input produces order-wise high-frequency features $\mathbf{Z}_h^{(m,r,q)} = \mathcal{H}_q(\mathbf{H}^{(m,r)})$. With $(\beta_1, \beta_2) = \text{softmax}(\mathbf{w}_\beta)$,

$$\mathbf{U}_{\text{wm}}^{(m,r)} = \text{LN}\left([\beta_1 \mathbf{Z}_h^{(m,r,1)} \parallel \beta_2 \mathbf{Z}_h^{(m,r,2)}] \mathbf{W}_o + \mathbf{H}^{(m,r)}\right), \quad (9)$$

where $\mathbf{W}_o \in \mathbb{R}^{2d \times d}$. Not every observed closure is clustering-relevant, so WMSC injects only the component of $\mathbf{U}_{\text{wm}}^{(m,r)}$ that is unexplained by the pairwise anchor $\mathbf{G}_h^{(m,r)}$. Let $\mathbf{g}_i$ and $\mathbf{u}_i$ denote the i -th rows of a pairwise anchor $\mathbf{G}$ and a closure field $\mathbf{U}$. With a routing map $\mathcal{R}$, the shared corroboration operator is

$$\begin{aligned} \text{Cor}(\mathbf{G}, \mathbf{U}; \mathcal{R}) &= \mathbf{G} + \eta \Pi_{\mathbf{G}}^\perp \mathcal{R}(\mathbf{U}), \\ (\Pi_{\mathbf{G}}^\perp \mathbf{u}_i) &= \mathbf{u}_i - \frac{\mathbf{u}_i^\top \mathbf{g}_i}{\|\mathbf{g}_i\|_2^2 + \epsilon} \mathbf{g}_i, \end{aligned} \quad (10)$$

with bounded scale $\eta = \eta_{\max} \sigma(s_\eta)$. WMSC returns $\tilde{\mathbf{Z}}_h^{(m,r)} = \text{Cor}(\mathbf{G}_h^{(m,r)}, \mathbf{U}_{\text{wm}}^{(m,r)}; \text{Id})$, so closure evidence supplements rather than overwrites the pairwise high-frequency geometry. Removing WMSC keeps only $\mathbf{G}_h^{(m,r)}$; dropping order 2 retains solely $\mathbf{Z}_h^{(m,r,1)}$ in Eq. 9; replacing $\Pi_{\mathbf{G}}^\perp$ by the identity injects $\mathbf{U}_{\text{wm}}^{(m,r)}$ directly onto $\mathbf{G}_h^{(m,r)}$.

2.4 Relation-Frequency Decoupled Calibration (RFDC)

WMSC still leaves a relation index on each view. A single global weight cannot express that one relation may stabilize interiors while another sharpens boundaries, so RFDC aggregates the low- and high-frequency channels separately. Let $\mathcal{E}^{(r)}$ be the edge set of $\mathbf{A}^{(r)}$, $e_r = |\mathcal{E}^{(r)}|$, $u_r = \log(1 + e_r)$, and $\bar{u}, \sigma_u$ the mean and standard deviation of $\{u_r\}_{r=1}^R$. For modality m and channel $b \in \{\ell, h\}$,

$$\begin{aligned} v_r &= \frac{u_r - \bar{u}}{\sigma_u + \epsilon}, \quad g_V = \mathbb{I}\left[\frac{\max_r e_r}{\min_{r: e_r > 0} e_r} \geq \kappa_V\right], \\ \omega_r^b &= (1 - g_V) \frac{1}{R} + g_V \text{softmax}_r\left(\frac{\ell_r^b + \gamma_b v_r}{\tau_{\text{rel}}}\right), \\ \mathbf{Z}_\ell^{(m)} &= \sum_r \omega_r^\ell \mathbf{Z}_\ell^{(m,r)}, \quad \mathbf{Z}_h^{(m)} = \sum_r \omega_r^h \tilde{\mathbf{Z}}_h^{(m,r)}, \end{aligned} \quad (11)$$

where ℓ_r^ℓ and ℓ_r^h are independent logits. Uniform fusion is retained under comparable volumes; severe imbalance activates volume-aware, frequency-specific weights, and when $R = 1$ no relation reweighting is applied.

2.5 Cross-modal Closure Corroboration (CMCC)

RFDC removes the relation index but still leaves modalities separate. Representation-level fusion can reweight unimodal features, yet it does not construct closures that appear

only when edges from different modalities are taken jointly. CMCC fills this gap before hierarchical fusion, reusing the shared operator in Eq. 10.

Let $\mathcal{T}(\mathcal{E})$ denote triangles induced by an edge set. CMCC keeps closures present in the multimodal union but absent from every unimodal triangle set:

$$\mathcal{T}_{\text{cm}} = \mathcal{T}\left(\bigcup_{m=1}^M \mathcal{E}_h^{(m)}\right) \setminus \bigcup_{m=1}^M \mathcal{T}(\mathcal{E}_h^{(m)}). \quad (12)$$

No new pairwise edge is invented. For each retained closure τ and edge $e = (u, v) \subset \tau$, provenance $a_{\tau,e}^m \in \{0, 1\}$ records whether e occurs in modality m . From the RFDC-aggregated order-1 highs $\mathbf{Z}_h^{(m,1)} = \sum_r \omega_r^h \mathbf{Z}_h^{(m,r,1)}$, write $\mathbf{z}_{h,i}^{(m,1)}$ for the i -th row. Edge and closure tokens are

$$\begin{aligned} \mathbf{d}_{\tau,e} &= \frac{\sum_m a_{\tau,e}^m \text{abs}(\mathbf{z}_{h,u}^{(m,1)} - \mathbf{z}_{h,v}^{(m,1)})}{\sum_m a_{\tau,e}^m}, \\ \mathbf{c}_\tau &= \frac{1}{3} \sum_{e \in \tau} \mathbf{d}_{\tau,e}. \end{aligned} \quad (13)$$

Averaging incident tokens at node i yields $\mathbf{f}_i$; stacking all nodes gives $\mathbf{F}_{\text{cm}} \in \mathbb{R}^{N \times d}$. With provenance deficit $\delta_\tau^m = 3 - \sum_{e \in \tau} a_{\tau,e}^m$, normalized weight $\rho_\tau^m = \delta_\tau^m / \sum_{m'} \delta_\tau^{m'}$, node-level mean $\bar{\rho}_i^m$, coverage χ_i , and routing $\mathcal{R}_{\text{cm}}^{(m)}(\mathbf{f}_i) = \chi_i \bar{\rho}_i^m \mathbf{f}_i$,

$$\hat{\mathbf{Z}}_h^{(m)} = \text{Cor}(\mathbf{Z}_h^{(m)}, \mathbf{F}_{\text{cm}}; \mathcal{R}_{\text{cm}}^{(m)}). \quad (14)$$

Cross-modal closure evidence is therefore injected as a residual on the RFDC high-frequency anchors, preferentially returning signal to modalities that contribute fewer closure edges. Disabling CMCC uses $\mathbf{Z}_h^{(m)}$ in place of $\hat{\mathbf{Z}}_h^{(m)}$; replacing $\mathcal{R}_{\text{cm}}^{(m)}$ by the identity removes provenance-guided routing; omitting the set difference in Eq. 12 retains unimodal closures in the union.

2.6 Hierarchical Fusion and Learning Objective

Frequency fusion combines the shared low-frequency interior with the corroborated high-frequency representation,

$$\mathbf{Z}^{(m)} = (1 - a_m) \mathbf{Z}_\ell^{(m)} + a_m \hat{\mathbf{Z}}_h^{(m)}, \quad a_m = \sigma(s_m). \quad (15)$$

When $M = 1$, CMCC is inactive and Eq. 15 uses the RFDC high branch $\mathbf{Z}_h^{(1)}$ in place of $\hat{\mathbf{Z}}_h^{(1)}$. Modality attention with weights ξ_m returns $\mathbf{Z} = \text{RowNorm}(\sum_m \xi_m \mathbf{Z}^{(m)})$.

Training combines scaled cosine reconstruction $\mathcal{L}_{\text{rec}}$, CPC InfoNCE terms $\mathcal{L}_\ell, \mathcal{L}_h$ (van den Oord, Li, and Vinyals 2018), optional modality alignment $\mathcal{L}_{\text{mod}}$, and DEC sharpening $\mathcal{L}_{\text{clu}} = \text{KL}(P \| Q)$ (Xie, Girshick, and Farhadi 2016):

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{clu}} + \lambda(\mathcal{L}_\ell + \mathcal{L}_h) + \mu \mathcal{L}_{\text{mod}}, \quad (16)$$

with $\mathcal{L}_{\text{mod}}$ disabled when $M = 1$. Labels are unused during representation learning; at inference, K -means clusters $\mathbf{Z}$ using the known cluster count K .

Methods	IMDB			Amazon			Ele-Fashion		
	ACC%	NMI%	ARI%	ACC%	NMI%	ARI%	ACC%	NMI%	ARI%
KMeans-Text	50.61	12.28	14.58	61.19	15.29	21.36	35.97	34.84	25.48
KMeans-Image	48.83	<u>14.34</u>	13.61	53.48	19.25	5.97	37.88	39.85	22.64
HDMI (WWW'21)	50.00	11.41	11.92	71.99	32.72	33.49	40.11	41.88	29.16
InfoMGF-RA (IJCAI'24)	51.45	12.83	15.22	72.40	33.06	34.81	40.30	47.36	30.83
GWMAC (CIKM'22)	42.52	4.02	8.03	51.48	21.68	19.39	33.19	39.83	22.25
GCFagg (CVPR'23)	53.20	13.43	<u>16.79</u>	69.80	29.08	32.12	39.14	51.69	30.38
MHGAT (NeuralComput.Appl.'22)	43.95	12.54	13.57	64.22	19.35	15.95	41.98	30.68	19.74
UniGraph2 (WWW'25)	47.16	6.44	5.07	64.19	13.18	11.52	31.21	19.36	16.70
DMGC (ICLR'24)	<u>56.33</u>	13.58	15.70	<u>79.12</u>	<u>34.62</u>	<u>46.31</u>	53.05	54.63	49.28
DGF (AAAI'26)	54.30	13.87	14.36	63.92	21.70	26.70	60.71	56.67	62.04
TopoCor	62.16	17.60	20.26	80.71	39.50	50.13	63.48	58.81	64.63

Table 1: Clustering performance comparison on multimodal graph datasets.

Methods	ACM			DBLP			Yelp		
	ACC%	NMI%	ARI%	ACC%	NMI%	ARI%	ACC%	NMI%	ARI%
VGAE (NeurIPS'16)	73.58	45.07	43.47	84.48	61.79	65.56	65.07	39.19	42.57
DGI (ICLR'18)	81.14	57.79	51.74	86.88	65.59	70.35	65.29	39.42	42.62
HDMI (WWW'21)	82.58	59.02	54.72	87.25	64.85	70.85	79.56	60.81	59.35
DMG (ICML'23)	87.96	63.41	67.26	88.45	69.03	73.07	88.26	65.66	66.33
BTGF (AAAI'24)	88.53	64.83	67.76	88.05	66.28	72.47	91.39	69.97	73.53
InfoMGF-RA (NeurIPS'24)	<u>90.63</u>	<u>68.11</u>	<u>73.07</u>	88.72	70.19	73.49	91.85	<u>72.67</u>	74.66
BMGC (ACM MM'24)	84.08	61.51	57.51	<u>90.84</u>	<u>72.27</u>	<u>78.05</u>	91.97	72.01	<u>75.18</u>
TopoCor	92.10	71.90	76.83	91.95	75.56	80.67	93.46	75.36	79.41

Table 2: Clustering performance comparison on graphs with multiple relations.

3 Experiments

3.1 Experimental Setup

We evaluate TopoCor on six datasets: multimodal IMDB, Amazon, and Ele-Fashion (Jia et al. 2023; Zhu et al. 2025), and multi-relational ACM, DBLP, and Yelp (Fu et al. 2020; Yun et al. 2019; Lu et al. 2019). Baselines include raw-modality K-means, multimodal and multi-relational graph encoders, and recent filtering methods (Jing, Park, and Tong 2021; Shen, Wang, and Kang 2024; Gong, Nie, and Xu 2022; Yan et al. 2023; Jia et al. 2023; He et al. 2025; Guo et al. 2025; Zheng et al. 2026; Kipf and Welling 2016; Velićković et al. 2019; Mo et al. 2023; Qian, Li, and Kang 2024; Shen, He, and Kang 2024). Clustering is measured by ACC, NMI, and ARI (Kang et al. 2025); K -means is applied to the learned representation with known K , using the lowest-loss checkpoint selected over five seeds without label supervision.

3.2 Main Results

Multimodal clustering. Table 1 supports three observations. (1) *Complementarity of topology and modality.* Baselines using features only (KMeans-Text/Image), methods for multiple relations (HDMI, InfoMGF-RA), and multimodal methods without graph topology (GWMAC, GCFagg) remain less competitive. Their relative performance suggests

that topology and modality provide complementary information for clustering. (2) *Explicit closure context complements pairwise integration.* Multimodal graph encoders (MHGAT, UniGraph2) couple topology with attributes, while graph filtering methods (DMGC, DGF) further refine frequency structure or suppress feature noise and form the strongest prior family. These methods operate primarily at the edge or representation level. TopoCor’s advantage over them is consistent with the value of explicitly modeling closure context. (3) *Effectiveness of topological corroboration.* TopoCor ranks first across all three multimodal benchmarks, consistent with corroborating pairwise high frequency discrepancies through closures within and across modalities rather than pairwise integration alone.

Clustering with multiple relations. Table 2 supports two observations. (1) *Integration across relations matters.* Single graph methods (VGAE, DGI) are consistently less competitive than methods that jointly model relation views (DMG, BTGF, InfoMGF-RA, BMGC), supporting the value of relation complementarity. (2) *Relation utility depends on frequency.* Strong fusion methods (BTGF, InfoMGF-RA, BMGC) integrate all relations but do not distinguish their roles in smooth cluster interiors and non-smooth boundaries. TopoCor’s advantage, together with the opposite low and

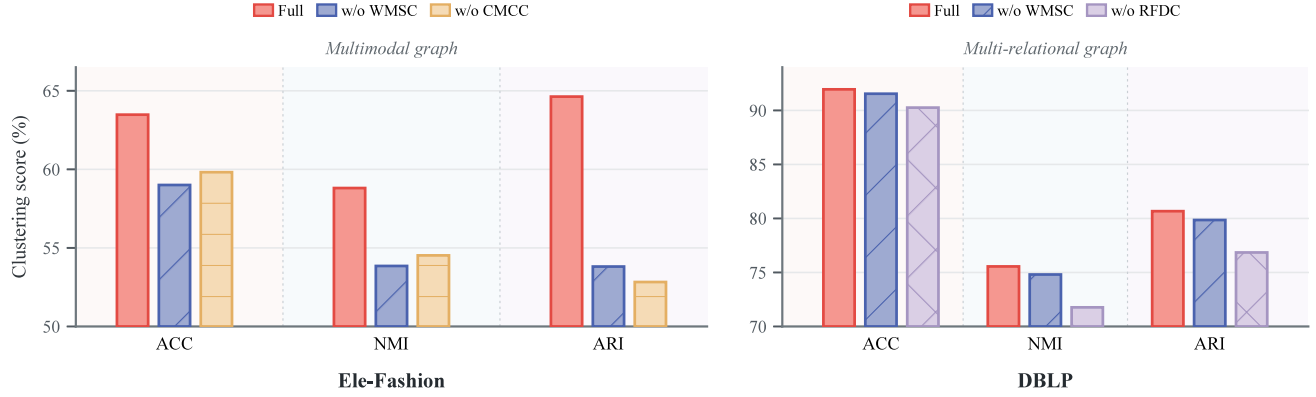


Figure 3: Component ablations on Ele-Fashion and DBLP. Bars report ACC, NMI, and ARI (%); patterned bars denote targeted removals of WMSC, CMCC, and RFDC.

WMSC configuration	ACC	NMI	ARI
Full TopoCor	63.48	58.81	64.63
w/o WMSC	59.00	53.85	53.81
w/o order 2	62.42	56.74	62.53
w/o orthogonal residual	61.71	54.72	55.42

Table 3: WMSC decomposition on Ele-Fashion (%).

CMCC configuration	ACC	NMI	ARI
Full TopoCor	63.48	58.81	64.63
w/o CMCC	59.82	54.52	52.83
w/o routing guided by provenance	61.70	54.72	55.43
w/o unimodal closure exclusion	62.72	55.73	60.62

Table 4: CMCC decomposition on Ele-Fashion (%).

high frequency relation preferences in Figure 5, supports relation calibration that depends on frequency while retaining corroboration based on closure.

3.3 Ablation Study

Specialization across input regimes. Figure 3 examines whether TopoCor activates the mechanisms warranted by each input structure. On multimodal Ele-Fashion, removing either WMSC or CMCC causes broad degradation. On single feature type DBLP, CMCC is inapplicable and the Relation-Frequency calibration path becomes decisive, while WMSC remains complementary. The modules therefore divide responsibility according to modality and relation structure rather than contributing uniformly across settings.

Within modality evidence requires interaction order and anchor preservation. Table 3 shows that removing WMSC degrades all three metrics. Retaining only order 1 evidence also regresses, demonstrating that triangle incidence contributes information beyond edge filtering. Direct closure injection is substantially less effective than the orthogonal residual update, indicating that higher order evidence is most useful when it contextualizes rather than overwrites the pairwise anchor.

Evidence across modalities must be novel and attributable. Table 4 shows that removing CMCC degrades all three metrics. Uniform routing weakens the result, demonstrating that edge provenance is necessary for assigning evidence across modalities. Retaining every union closure is

also inferior, indicating that topology already present within one modality is largely redundant. The gain therefore comes from identifying jointly supported closures and routing their evidence, rather than merely adding more triangles.

3.4 Analysis

Representation geometry reflects partition coherence. Figure 4 compares the learned geometry on Amazon. UniGraph2 and DGF exhibit substantial class mixing, while DMGC recovers coarse group structure. TopoCor forms the most compact neighborhoods consistent with class labels and attains the highest Silhouette coefficient. The reduced overlap is consistent with closure corroboration organizing pairwise high frequency variation into coherent cluster structure.

Why relations should be fused by frequency. Figure 5 shows that a relation does not have one fixed importance. On DBLP, the sparse Relation 1 contributes more to the low frequency channel, whereas the dense Relation 2 contributes more to the high frequency channel. Uniform averaging ignores this difference, and a shared gate still forces both channels to prefer the same relation. RFDC instead learns separate weights for low and high frequencies and corrects for relation size, so each relation is used according to the information it provides in each frequency band.

Corroboration stabilizes neighborhood expansion. Figure 6 enlarges the homophilic and heterophilic neighborhoods, progressively admitting edges with lower confidence. DMGC peaks at narrow neighborhoods and deteriorates as

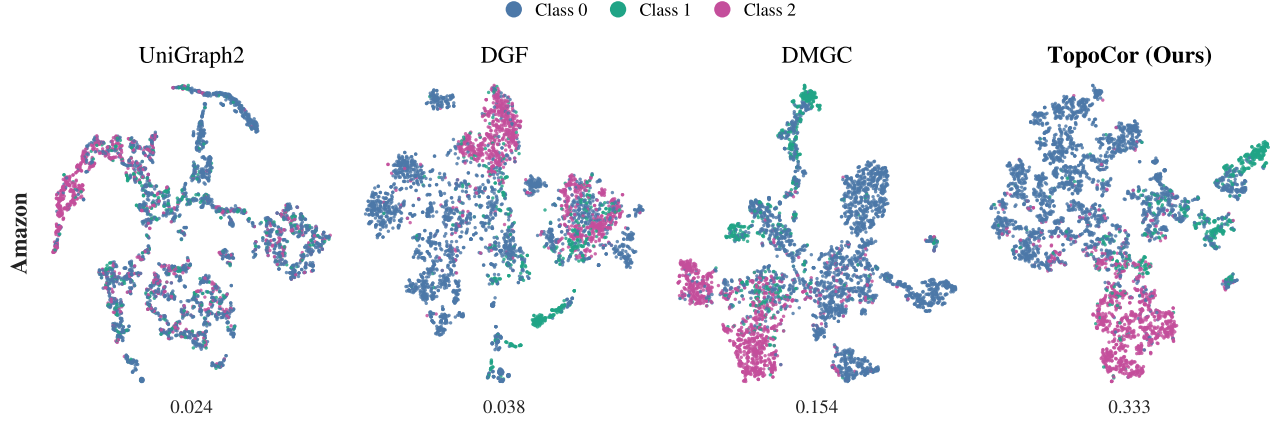


Figure 4: t-SNE visualization of learned node representations on Amazon.

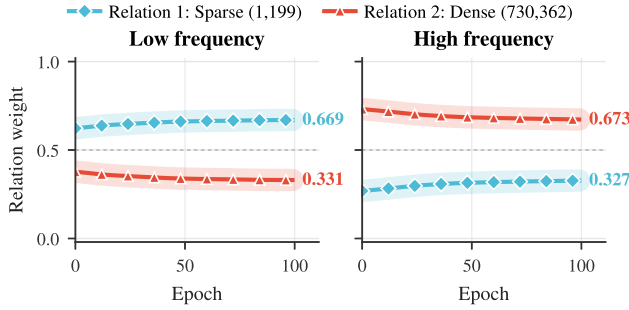


Figure 5: Relation-Frequency weights on DBLP, where Relation 1 has 1,199 edges and Relation 2 has 730,362 edges. The low and high frequency channels favor opposite relations.

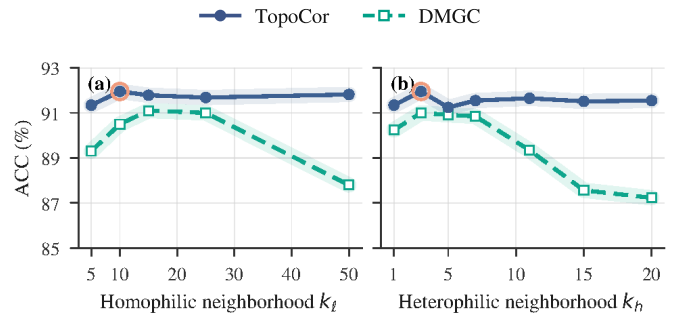


Figure 6: Sensitivity to neighborhood sizes of the homophilic and heterophilic graphs on DBLP.

either support grows, whereas TopoCor remains stable over a broader range.

4 Related Work

Multimodal graph clustering with multiple relations. Multimodal graph clustering couples attributes and topology through encoding, frequency decomposition, denoising, or alignment (Guo et al. 2025; Zheng et al. 2026; Wang et al. 2025a; Jia et al. 2023; He et al. 2025; Gong, Nie, and Xu 2022; Yan et al. 2023; Xu et al. 2022; Zhang et al. 2019), while methods for multiple relations learn complementary relation views (Shen, Wang, and Kang 2024; Shen, He, and Kang 2024; Mo et al. 2023; Qian, Li, and Kang 2024; Jing, Park, and Tong 2021). Frequency-aware models improve cluster interiors by separating smooth and non-smooth channels, and multimodal encoders further exploit visual-textual complementarity. Despite these advances, modalities are typically fused after separate encoding and each relation receives a single global weight, so closures across modalities and frequency-dependent relation roles remain open.

Higher order and heterophilic graph learning. Higher order models expose organization beyond pairwise edges (Benson, Gleich, and Leskovec 2016; Chodrow, Veldt,

and Benson 2021; Bodnar et al. 2021; Benson et al. 2018; Feng et al. 2019), and heterophilic learners employ specialized propagation or contrastive objectives (Wang et al. 2025b; Li et al. 2025, 2026b,a; Zhu et al. 2020; Luan et al. 2022; Li et al. 2022), mainly for supervised unimodal classification where labels can guide the use of higher order structure. Unsupervised multimodal clustering instead observes candidate heterophily only as pairwise high frequency variation. TopoCor addresses this gap by corroborating discrepant edges with observed closures within and across modalities, instead of relying on deeper pairwise message passing.

5 Conclusion

We presented TopoCor, which corroborates pairwise heterophilic evidence with observed simplicial closures within and across modalities, and calibrates multi-relational inputs in a frequency-specific manner. By preserving low frequency interiors while contextualizing high frequency discrepancies through WMSC, RFDC, and CMCC, it shifts heterophily modeling from pairwise detection to topological corroboration. Across six benchmarks, including the nearly 100k-node Ele-Fashion graph, TopoCor improves clustering over strong multimodal and multi-relational baselines.

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